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MA250 – Data Visualization

Final Project

Introduction

The vibrant and diverse boroughs of New York City attract millions of visitors each year, whether they come for business or tourism, and the city is an international icon. With so many visitors, short-term rental platforms like Airbnb have become crucial components of the city's hospitality industry. This project uses a comprehensive dataset to explore key factors which impact Airbnb listings in New York, and from this data we hope to answer three primary questions. First, how does the average price of an Airbnb rental vary by neighborhood and borough? Understanding price trends can help hosts optimize their pricing strategies and help consumers find accommodations that fit their budget. We also want to investigate how listing availability relates to the price, which may shed light on supply and demand patterns. Finally, we want to investigate which variables are most related to the listing review score. These research questions are relevant to both hosts and guests, permitting hosts to prioritize what is important and price accurately, but more importantly provide additional information to guests who want to stay in the city.

Dataset Overview

This data was provided by Airbnb directly, and is a snapshot of Airbnb activity in New York City as of January 2024, including over 20,000 listings, outlining geographic data, host information, listing details, and performance metrics. The geographic data contains boroughs, specific neighborhoods, and actual longitudes and latitudes, which offer multiple different layers of analysis. The host information includes ID, name, and count of listings which may provide insights between large-scale hosts and individuals with fewer listings, and possibly the types of properties they tend to list. Listing characteristics include variables such as room_type, bedrooms, and bathroom counts. Performance metrics include ratings and availability, which reveal how popular each listing is. Little cleaning/changes were required other than changing variable types and creating calculated fields in Tableau. Our exploratory data analysis revealed that the price variable has a strong positive skew, with a low median and mode but with a high mean, being pulled up by high-end Airbnb listings, as seen in Figure 1 below. Additionally, the ratings variable which tracked review scores was negatively skewed as shown in Figure 2, signaling that high review scores are very important to success and a review score of 4.0 may not be as good as it sounds.

Figure 1:

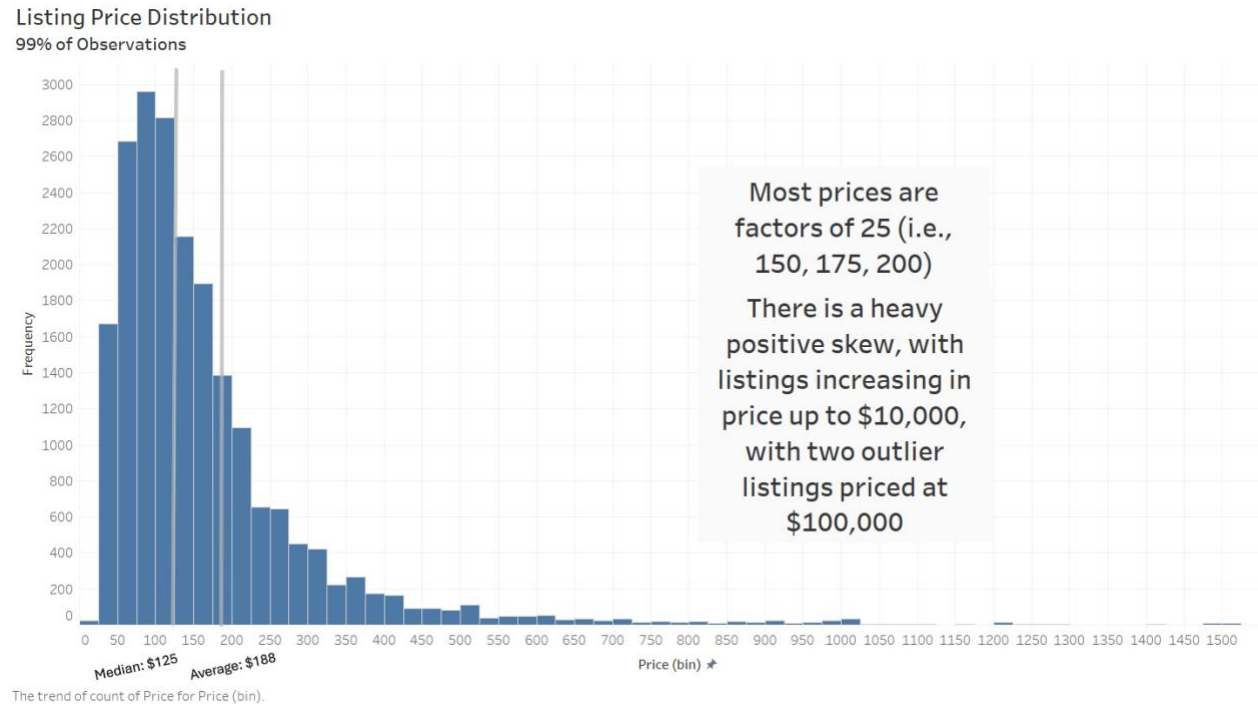
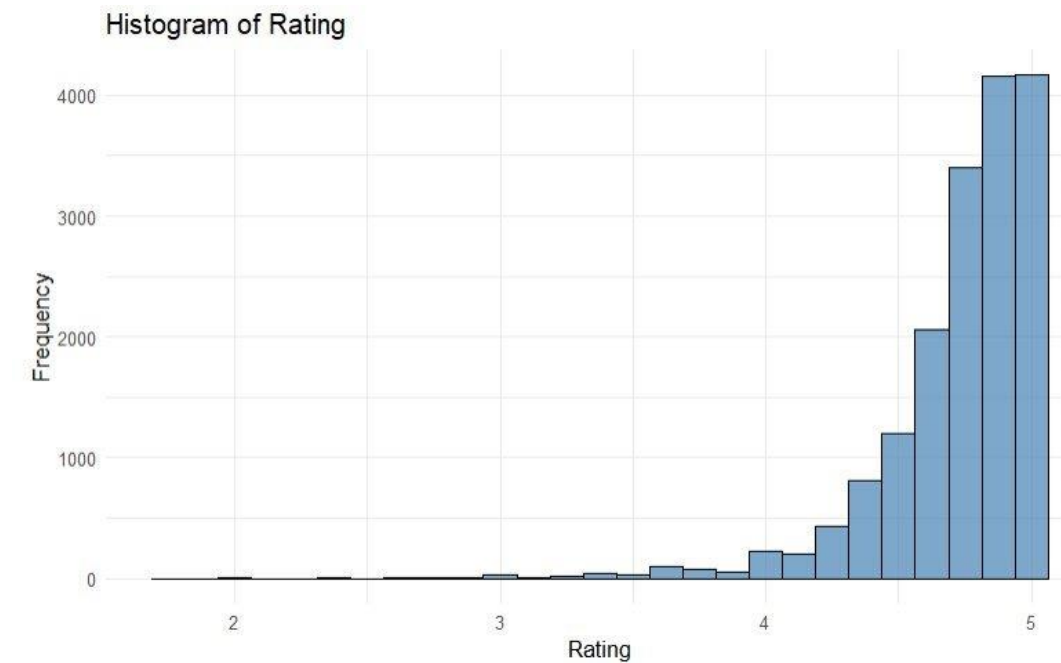


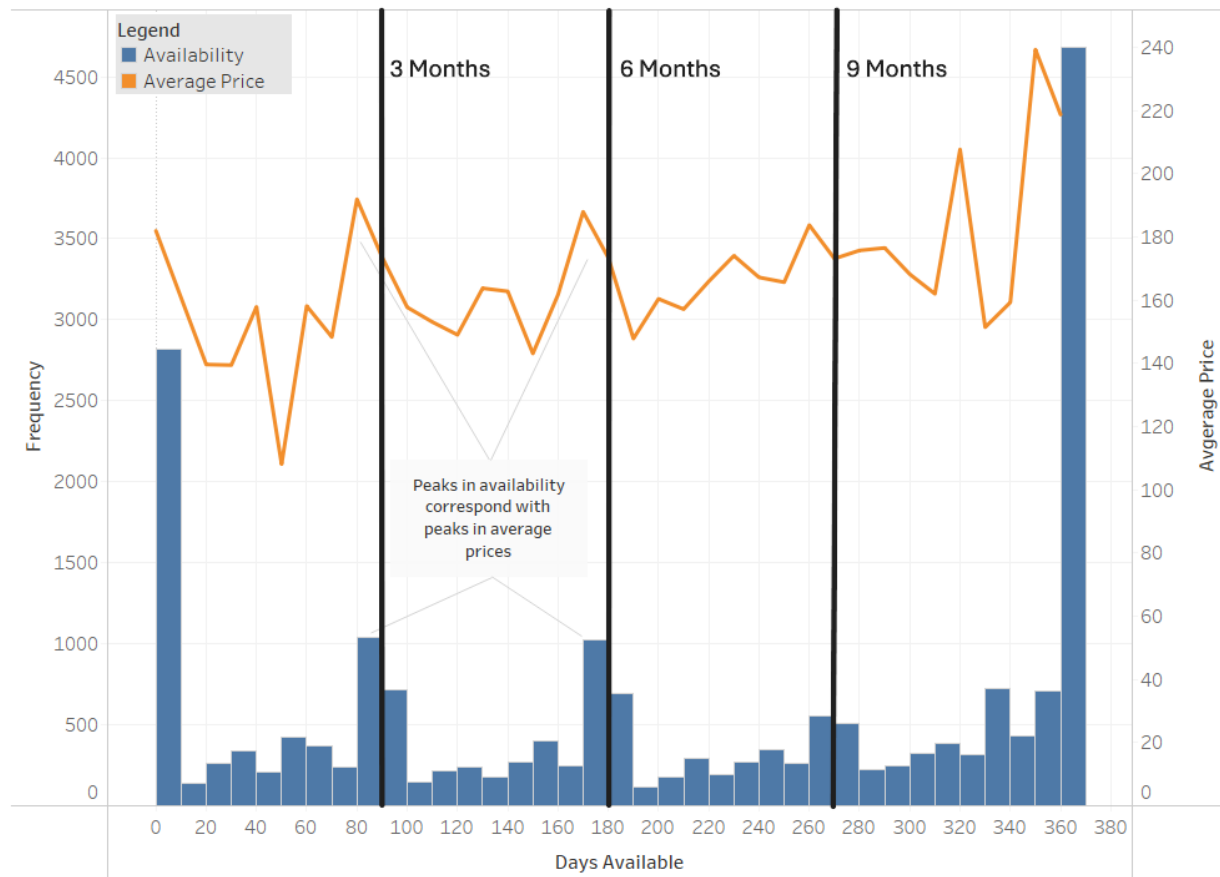
Figure 2:



Visualizations and Analysis

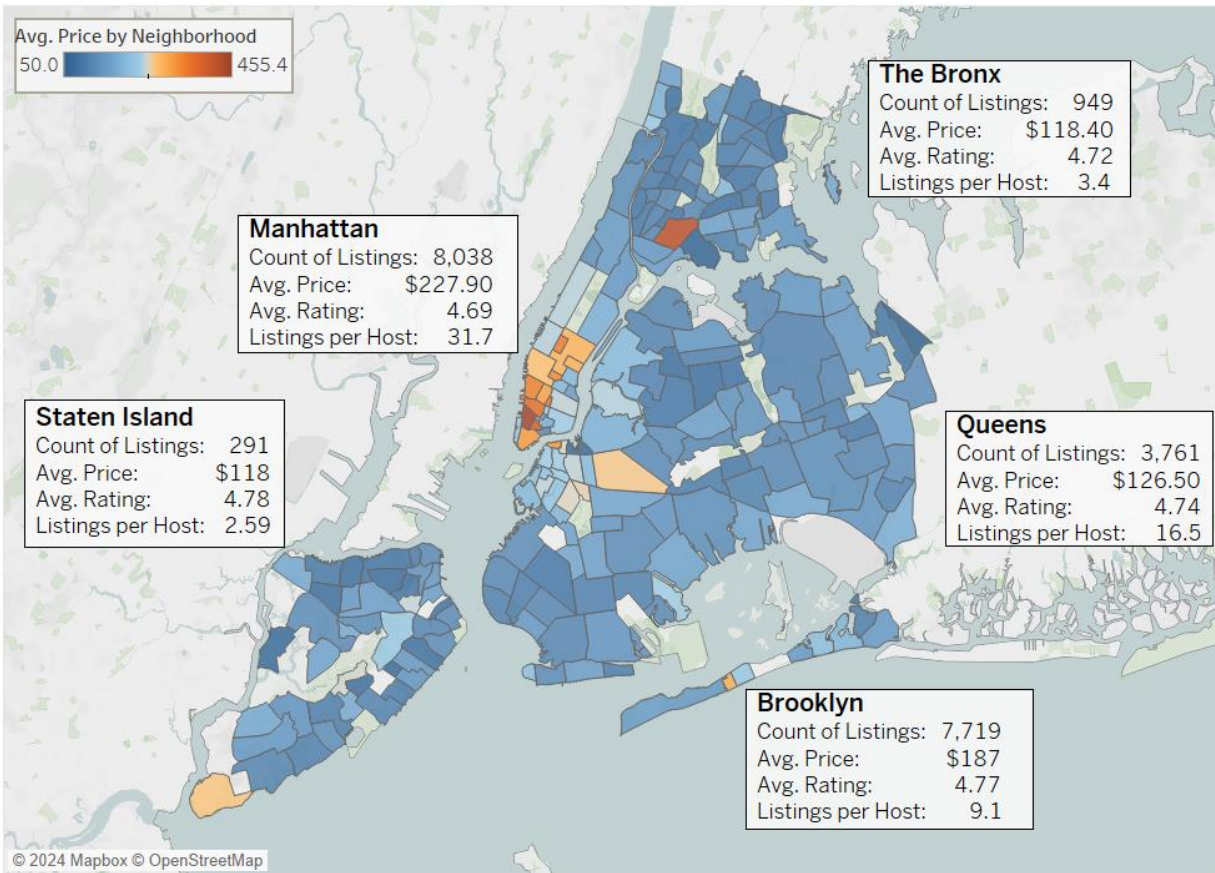
Despite low correlation, there is a relationship between Price and Availability

Correlation of 0.02 shows lack of consistent relationship between the variables, but there is an underlying pattern

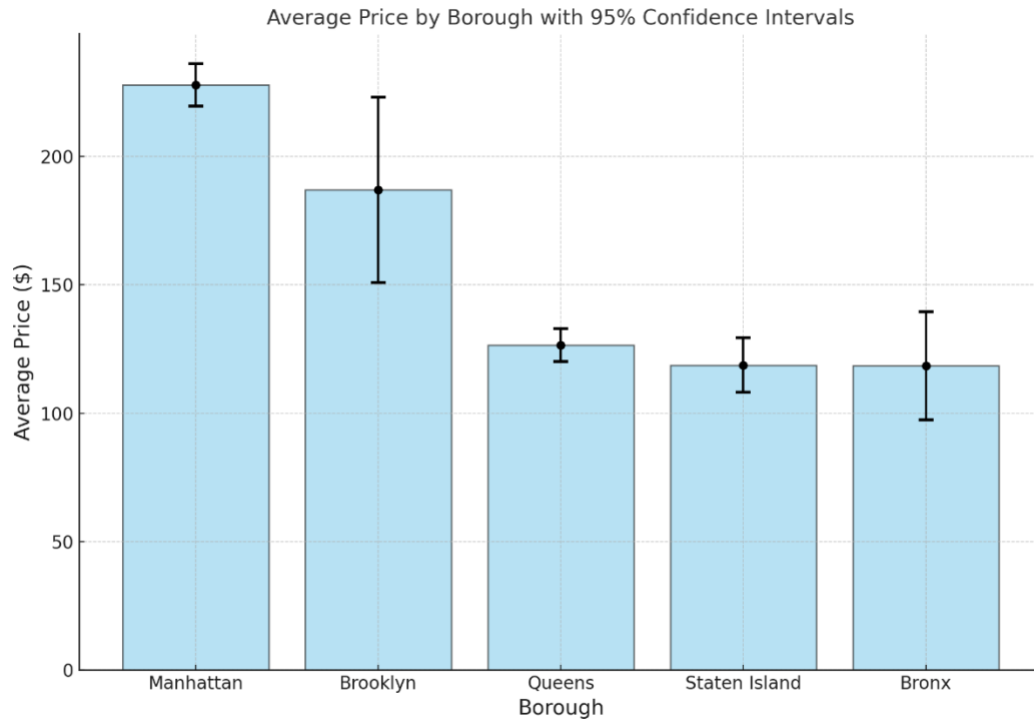


This chart shows the correlation between price and availability, with the orange line representing the average price for those listings, and the blue bars representing the frequency of each listing corresponding to the days available. Although statistically there is not much of a correlation, peaks in availability often align with peaks in average prices, suggesting there is an underlying pattern. Hosts are able to list their property available for partial parts of the year, and similar to high availability frequency at 0 and 365, this pattern continues in intervals of 3 months, as designated by the black lines. This relates to our research question about supply and demand for Airbnb, which can explain how the hosts' preferences affect price.

Manhattan has the highest average pricepoints and most listings overall
Staten Island has the least listings and nearly ties with **The Bronx** for lowest price



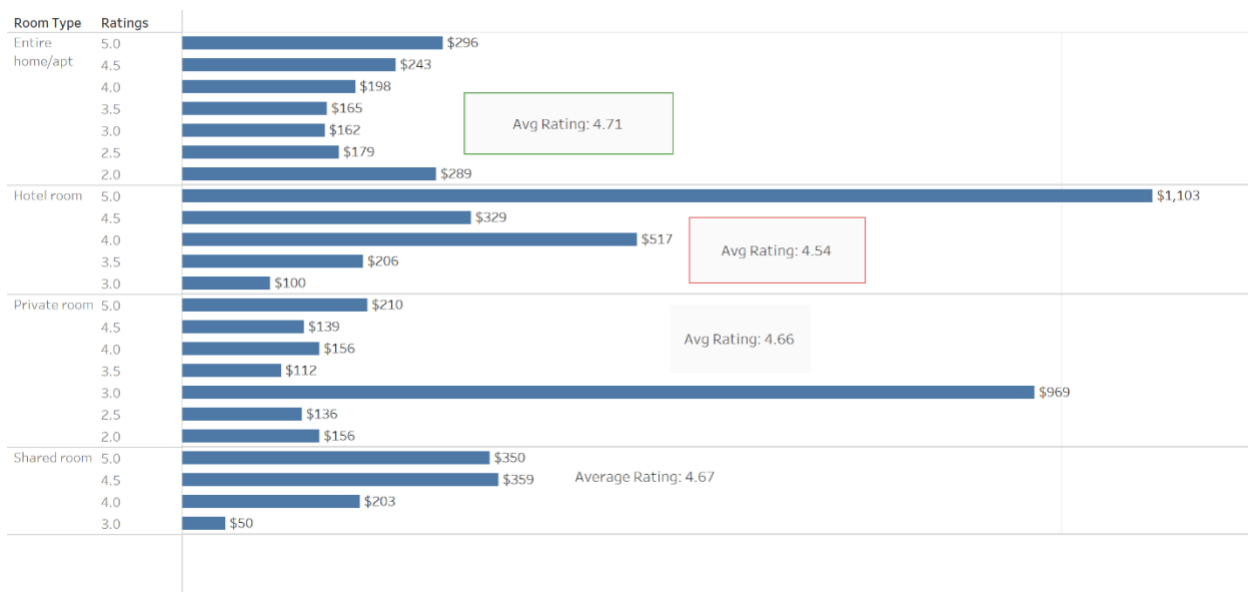
To answer our research question about how prices vary by borough and neighborhood, we made a choropleth map. This map shows the average price by neighborhood with the color representing price. The darker blue colors show the cheapest priced neighborhoods, while the darker red color shows the most expensive. Additionally, it is annotated with statistics about the boroughs, including number of listings, average price and rating, and listings per host. This graphic relates to our question by showing that lower Manhattan has some of the most expensive neighborhoods in NYC.



Brooklyn has a wide interval with some listings similar to Manhattan, while Queens, Staten Island, and the Bronx are similar to each other.

In order to better understand the distribution of average prices by borough, we made a bar chart with 95% confidence intervals to highlight the variability in price. This will give us a better understanding of how accurate our price estimates are and provides a range to see the true averages. Brooklyn has a wide confidence interval, suggesting that there are very inexpensive and very expensive listings in that borough. On the other hand, Manhattan has the highest prices and a relatively narrow confidence interval, indicating more consistent pricing.

On a Budget? Room Types in Manhattan by Rating and Price



Lastly, a clustered bar chart shows the average price of Airbnb listings in Manhattan categorized by room type and rating. Additionally, an annotation shows the overall average rating of each room type, with the green highlighted box (entire home/apartment) being the highest rated, and the red being the lowest. We chose Manhattan because it is likely the location where our audience would be visiting, and it had the greatest number of listings out of the 5 boroughs. The 4 room types available are entire homes/apartments, hotel rooms, private rooms, and shared rooms. Private rooms appear to be the cheapest overall room type for listings with 4+ star ratings, however, entire homes/apartments are not much more expensive. Hotel rooms have the greatest variation and price and the lowest average rating, so visitors on a budget should avoid them.

Findings – summarize insights, how they answer our rq, broader implications of our results

A positive correlation between price and availability is present at critical intervals like full-year and short-term available listings. A potential reason for this is seasonal houses, for example beach houses. A beach house may be open in only the summer months (~90 days) or for every season except winter (~270 days). Because places on the water are generally more expensive due to their location, that could be causing the spikes in price for these specific intervals. The spike in price for short-term rentals could be due to places only available for major events, like sports championships or music festivals. Since these listings are available for specific times and reasons, they are generally going to be more expensive than your average rental. This can help explain the supply and demand dynamics from the different availabilities.

Manhattan is the most expensive of the 5 boroughs, but its specifically listings in lower Manhattan that drive up the price. Lower Manhattan is what people think of when they think 'NYC'. Central Park, The Empire State Building, The MET, Time Square, are all located in lower Manhattan, making it a hotspot for tourists. This increases demand, which drives up the prices. Additionally, the average listings per host in Manhattan is almost 32, which is much greater than the other boroughs. This could explain the lower rating, as a host with 32 listings is less likely to personalize the rental experience and engage with their customer. Staten Island, the borough with the least listings per host (2.6) has the highest average rating (4.78), which further supports this claim. Brooklyn and Manhattan dominated the Airbnb market in NYC, with 7.7k and 8k listings respectively, which also helps explain why they are the most expensive places to stay in. A visitor on a budget who wants the NYC experience could stay in central or upper Manhattan. They will be close to everything, while getting a good deal on their house/room. Brooklyn has a relatively diverse market of Airbnb's, reflected by its confidence interval. This indicates a mix of luxury and budget listings, which would help it cater to a broader range of visitors.

Lastly, a higher rating will most likely result in a higher priced listing. Private rooms seem to be the cheapest room type, as 4.5 and 5 star rated shared rooms are on average over \$300 a night. If visitors are seeking privacy, entire homes/apartments over 4 stars are still priced competitively, around \$250 a night. This reflects the premium placed on quality, as higher ratings often means better amenities, locations and overall rental experience. Ultimately, there are good options for any room type that suits the visitor's needs.

Conclusion and Future Decisions – potential next steps, further possible analysis

One next step is continuing the analysis on pricing and availability and looking at seasonal trends with time series analysis. Having the ability to break down the data based on season or higher demand periods would allow us to analyze if that is the actual cause of the price and availability pattern we noticed. Next, we could do case studies on some of the outliers, and potentially remove them from our data so certain statistics are not skewed. These outliers could be unserious listings, errors with accidental extra zeros, or host experiments with no intent on renting out the property yet. Removing these from our data will further increase our accuracy and prevent incorrect analysis. We can also do additional case studies on shared rooms and explain why they are so much more expensive than entire homes and private rooms.

Overall, Airbnb pricing is determined by a combination of availability, demand, location and quality. Seasonal and event specific listings most likely drive prices up for low availability, or 90/270 day availability listings, reflecting a greater demand during these times. Manhattan has the greatest number of listings out of the boroughs, and lower Manhattan is the most expensive

place to stay in NYC. Brooklyn also offers a lot of places, for a broader range of prices. Travelers looking for privacy and quality will find value in highly rated entire homes/apartments, and budget-conscious visitors can find deals with private rooms in upper Manhattan or Brooklyn to stay close to the attractions.